

Dynamic LAB Consortium Models for Plant-Based Fermentation

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Abstract. The global growth in plant-based food consumption has made chickpea and fava bean strong candidates for replacing animal protein due to their nutritional profile. However, the presence of *off-flavors* - including herbaceous, bitter, and beany notes - has limited their acceptance among consumers. These sensory attributes arise from complex biochemical pathways, including lipid oxidation mediated by lipoxygenase and protein degradation. Traditional empirical screening is insufficient to address this challenge, as it overlooks the metabolic dynamics of multi-strain communities. This work proposes the use of Genome-Scale Metabolic Models (GEMs) of selected lactic acid bacteria (LAB), analyzed by Flux Balance Analysis (FBA), to computationally model the behavior of microbial consortia in legume matrices. Using community modeling tools such as COBRApy, SMETANA, and MICOM, different simulation approaches will be evaluated with the aim of predicting the mitigation of *off-flavors* and identifying optimized simulation conditions.

Keywords: Biological Modeling · Lactic Acid Bacteria · Microbial Consortia · Genome-Scale Metabolic Models · Fermentation.

1 Introduction and Objectives

In recent years, plant-based foods have experienced remarkable growth globally, driven by nutritional, environmental and ethical concerns related to animal welfare [1,2]. Legumes, particularly chickpea (*Cicer arietinum*) and fava bean (*Vicia faba*), play a crucial role in this context, being rich sources of vegetable protein, fiber, vitamins and minerals, making them excellent candidates for replacing animal protein [3,4]. However, their consumer acceptance is hindered by a persistent sensory obstacle: the presence of undesirable aromas collectively referred to as *off-flavors*, including beany and grassy notes, bitterness, and earthy aromas [5]. These attributes result from three main biochemical pathways: lipoxygenase-mediated lipid oxidation, which originates volatile aldehydes such as hexanal and (E)-2-nonenal [6,7]; protein degradation and amino acid catabolism through the Ehrlich pathway and Strecker degradation [8,9]; and the accumulation of other volatile compounds from the Maillard reaction [10].

The conventional approach to mitigate these compounds relies on empirical screening, however this presents significant limitations, such as high costs, biological and experimental variability and a lack of mechanistic understanding [10,11]. These limitations are particularly relevant in multi-strain microbial communities, where interactions such as cross-feeding and substrate competition shape fermentation outcomes in ways that cannot be predicted from single-strain behavior [12]. Although microbial consortia offer well-documented functional advantages, including greater stability and metabolic complementarity, their rational design remains a challenge [13]. In this context, genome-scale metabolic modeling emerges as a promising tool. GEMs allow the reconstruction of complete metabolic networks and the computational simulation of metabolic fluxes, interactions between microorganisms and metabolite production under different conditions [14,15]. The simulation of microbial consortia can follow different strategies, from steady-state approaches to dynamic models, depending on the biological question, the quality of available models and the available experimental data [12]. The present project aims to computationally model LAB consortia in chickpea and fava bean matrices in order to predict *off-flavor* mitigation and identify optimized fermentation conditions. Specifically, the following objectives are defined: (i) curate and standardise GEMs of 2–3 LAB strains in SBML format; (ii) define medium conditions representative of legume matrices in COBRApy; (iii) analyse potential metabolic interactions and cross-feeding in the consortium using SMETANA; (iv) implement community simulations with MICOM to predict metabolic fluxes and *off-flavor* mitigation.

2 State of Art

2.1 Off-flavors

The undesirable sensory profile of chickpeas and fava beans results from the presence of volatile compounds generated by biochemical pathways native to the plant [5,6]. The most studied pathway is lipid oxidation mediated by lipoxygenase (LOX), which catalyses the conversion of polyunsaturated fatty acids into hydroperoxides, subsequently cleaved into short to medium chain volatile aldehydes such as hexanal and (E)-2-nonenal, responsible for grassy and rancid notes [6,7]. Simple hydration or grinding of the legumes is sufficient to trigger this cascade, meaning that these *off-flavors* originate intrinsically within the plant [5]. Additionally, protein degradation and amino acid catabolism through the Ehrlich pathway and Strecker degradation generate compounds such as 3-methylbutanal and phenylacetaldehyde [6,8], while the Maillard reaction contributes sulphurous and roasted volatiles [10]. Heat treatments, even though they are efficient in inactivating LOX, compromise protein functionality and nutritional value [3], justifying the use of fermentation as an alternative strategy.

2.2 Legume Fermentation with LABs

Lactic fermentation, a widely explored strategy for optimizing the sensory properties of legumes [11,16], preserves the protein functionality and nutritional pro-

file of the matrix, constituting a technologically relevant and effective alternative [2,3].

LABs contribute to the mitigation of off-flavors through two main mechanisms: indirectly by producing lactic acid, which causes a drop in pH, inhibiting LOX, whose optimal activity occurs at a pH between 6.0 and 6.4 [7,11]; and directly through the enzymatic action of the LABs themselves, which convert undesirable aldehydes into alcohols and carboxylic acids [16].

Among the LAB species with greatest relevance and the most documented experimental evidence in legume fermentation, *Lactiplantibacillus plantarum* stands out for its efficient acidification capacity and extensive proteolytic system, with functional ADH enzymes directly associated with hexanal reduction in legume fermentations [11,17]. *Leuconostoc mesenteroides* and *Pediococcus pentosaceus* are equally relevant candidates, with demonstrated capacity to reduce volatile aldehydes in legume matrices [4,18]. Additionally, some LABs produce positive aroma compounds, which can contribute to masking residual *off-flavors* and enriching the final sensory profile [19].

2.3 Genome-Scale Metabolic Models

GEMs are computational tools that integrate genomic, biochemical and physiological data, establishing the link between genotype and phenotype [14,20]. These models aim to represent the totality of an organism’s metabolic reactions, allowing predictions of growth rates and metabolic behavior under different environmental conditions [15]. These predictions are typically obtained through Flux Balance Analysis (FBA), which formulates a linear programming problem: maximizing an objective function, usually biomass growth, subject to the condition of a semi-steady state and the flux limits of each reaction [20]. GEM construction follows an established pipeline: genome annotation, mapping reactions to reference databases such as KEGG and BiGG, manual curation of gene-protein-reaction relationships and experimental validation [20,21].

Model quality is critical for the reliability of predictions, as models with incomplete annotation produce unreliable results [12]. Recently, GEMs have been applied to the study of microbial consortia, allowing the modeling of interactions between organisms and the prediction of phenomena such as cross-feeding and substrate competition [12,22]. Curated GEMs exist for the candidate LABs: iLP728 for *L. plantarum* and iLME620 for *L. mesenteroides*, available in databases such as BiGG and VMH [17,18]. For *P. pentosaceus*, since no curated model is published, automated reconstruction via tools such as CarveMe or ModelSEED constitutes the most appropriate approach [23,24,25], although the quality of the resulting model needs to be rigorously evaluated.

2.4 Microbial Consortia and Computational Community Modeling

While monoculture fermentation offers greater experimental control, single-strain systems are limited in terms of metabolic potential [13]. Microbial consortia provide metabolic complementarity - products from one species serve as substrates

for another (cross-feeding) - allowing for a more effective reduction of hexanal and other aldehydes compared to monocultures [13,26]. However, the behavior of microbial consortia is extremely complex, which requires the use of advanced computational tools to predict community-level outcomes [12].

Scott et al.[12] conducted a structured analysis of GEM-based consortium modeling tools, comparing methodologies in terms of predictive accuracy, scalability, and ease of use. The main conclusions indicate that, for small consortia (2-4 species) with known high-quality models, steady-state approaches such as cFBA and SteadyCom present the best balance between accuracy and computational cost, with the quality of individual models being the most limiting factor in predictions.

The cFBA approach models the consortium under balanced growth conditions and can be implemented either via pooling - merging all species into a single shared metabolic network and treating the community as a single organism - or via compartmentalization, keeping individual models separate with explicit exchanges between compartments [12,13,27]. SteadyCom extends this approach by simultaneously optimizing the growth of all species. MICOM implements a two-level optimization approach - maximizing community growth first and then individual member growth - and is particularly suitable for predicting cross-feeding and stable relative abundances [12,21]. SMETANA performs static analysis of potential metabolic exchanges between pairs of species, calculating interaction scores that identify likely cross-feedings before running full simulations [22]. More complex approaches such as dynamic models (dFBA), game theory, or multi-objective optimization are described in the literature, but are outside the scope of this project [12]. Practical implementation typically relies on COBRApy for GEM manipulation and FBA application [20], SMETANA and MICOM for interaction analysis and community simulation [22]. The combined use of these tools allows for a progressive and computationally efficient approach - from the static analysis of potential interactions with SMETANA to community simulation with MICOM - suitable for the exploratory nature of this project [12,21].

3 Preliminary Analysis

The system to be modeled consists of a small consortium of 2-3 LABs for chickpea and fava bean fermentation, with the aim of predicting metabolic interactions, cross-feeding, and potential mitigation of *off-flavors*. The nutritional composition of the legumes, with emphasis on the compounds responsible for *off-flavors* - precursors of hexanal and (E)-2-nonenal, amino acids involved in the Ehrlich and Strecker pathways, and major nutrients - will be incorporated as medium constraints in the models, based on data from the literature. The main anticipated technical challenges include harmonizing the nomenclature of metabolites from different GEMs, mapping *off-flavors* compounds to identifiers in existing models, and choosing the most appropriate simulation methodology given the quality of available models. The methodological sequence comprises:

curation and standardization of the GEMs in SBML format; definition of the legume media in COBRApy; interaction analysis via SMETANA; and community simulations with MICOM. The primary tools are COBRApy, SMETANA, and MICOM. In all these stages, the model analysis will be based on FBA implemented in COBRApy, serving as a basis for comparing growth, important metabolite fluxes, and the impact of different media and consortium conditions.

4 Work Plan

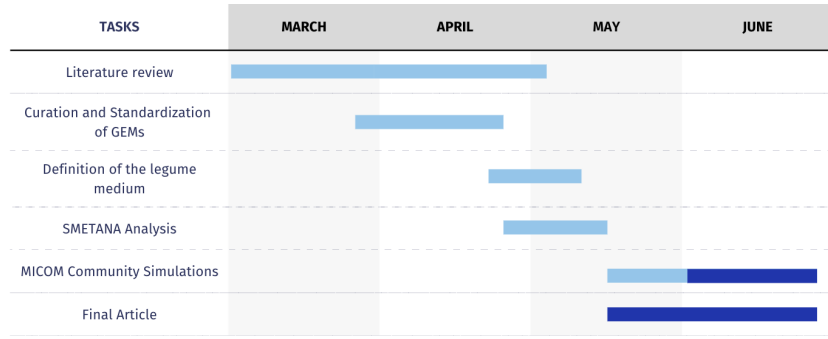


Fig. 1. Gantt chart representing the project work plan. Light blue bars represent the analysis and exploration phase and dark blue bars represent tasks for production and delivery of final results.

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